What we measure in Daily Trail, and what we do about it

Daily Trail is a daily treasure hunt for people who love puzzles. Everyone gets the same hunt each day: a riddle, a cipher and a pattern. The answers spell a three-letter treasure word. The game is working if players come back and finish the hunt week after week.

No real player data yet.

The game is live and already records what players do, such as opening the game, taking a hint, finishing and sharing. That data stays on each player's device for now. So this document is the plan, and nothing in it is a result yet.

THE BRIEF, AND WHERE EACH PART IS ANSWERED

"Jot down the metrics to be tracked on the game to share inferences to the management and to track what works and doesn't to improve the product."

The metrics to track	The main number and seven questions
What works, what doesn't, and how we improve the game	The table under each question
How findings reach management	The Monday update
Where the goals come from	How we set goals
How far to trust the numbers	Be careful when reading the numbers

THE MAIN NUMBER

Players who finish a hunt each week

The number of different players who finish at least one hunt in a week, Monday to Sunday. Opening the page only shows someone visited. Finishing shows they got what they came for. Counting by week also means one missed day does not matter.

How we set goals

Each goal comes from one of two places.

Published numbers, for players coming back

This is the only area where other games share reliable numbers, so these goals come from the table below.

Our own first two weeks, for everything else

We could not find reliable numbers from other games for things like finishing, hints, streaks or sharing. So we will measure them for two weeks first, and set goals from that.

Our first two weeks means 14 days of real data with at least 200 players in each group we measure. Fourteen days lets every daily hunt run twice.

Is a difference real? Small groups move around by chance. With 200 players, a number can move about 7 points up or down just by chance, and about 5 points with 400 players. In this document, more, fewer and much more all mean a difference bigger than that.

What other games achieve

Share of new players who	Typical puzzle game	Top quarter of all mobile games (our goal)	50 most successful puzzle games
Come back after 1 day	About 20%	**About 27%**	About 43%
Come back after 7 days	About 4.5%	**About 7.5%**	About 25%
Come back after 30 days	About 1.2%	**About 2.8%**	About 16%

We aim for the top quarter. That is our own choice: this game is for people who already love puzzles, so it should beat a typical game. We do not expect to match the 50 most successful puzzle games. They are shown for comparison.

These numbers come from phone apps.
Daily Trail is a website with no install and no notifications, so the comparison is close but not exact. We could not find a daily puzzle website that shares these numbers, and The New York Times does not release player numbers for its games.

Two columns use day 28.
That is the closest published number to day 30. The last column uses day 30.

The goals can change.
We will update them after our first two weeks if our players behave differently.

Seven questions we answer each week

Each question lists what we measure and the goal. The table under it shows what we do if a number looks wrong. We decide this in advance, so the numbers lead to action.

Do players come back?

Come back after 1 day

New players who open the game exactly 1 day after their first visit

About 27%

Come back after 7 days

New players who open the game exactly 7 days after their first visit

About 7.5%

Come back after 30 days

New players who open the game exactly 30 days after their first visit

About 2.8%

Hunts finished per player

Hunts finished in a week, divided by the players who finished at least one

Set after two weeks

What we see	What it probably means	What we do
Fewer than 20% of new players come back after 1 day	The first game gave them no reason to return	**Make tomorrow's hunt and the streak easier to see when they finish. Then try a daily reminder.**
20% or more come back after 1 day, but fewer than 4.5% after 7 days	The game gets boring, or players notice the puzzles repeat	**Add new puzzles first. Then try weekly goals.**
4.5% or more come back after 7 days, but fewer than 1.2% after 30 days	Regular players run out of reasons to stay	**Add longer-term goals, such as playing in teams or a monthly story map.**

Do they finish?

Players who start

Players who press Start, out of players who open the game and have not finished today

Set after two weeks

Players who finish

Players who finish, out of players who pressed Start that day

Set after two weeks

Players who quit at each stage

Players who see a stage but do not solve it, shown separately for the riddle, cipher and pattern

Compare the 3 stages

What we see	What it probably means	What we do
Fewer players press Start than in our first two weeks	The intro does not make people want to play	**Shorten the intro and show the first puzzle sooner.**
Fewer players finish than in our first two weeks, and most of them quit at one stage	That stage is the problem	**Fix that stage before writing new puzzles.**
Many more players quit at the cipher than at the riddle or pattern	The cipher instructions are too hard	**Make the instructions and the first hint easier.**

Are the puzzles right?

Similar puzzles means the same kind: riddles with riddles, ciphers with ciphers, and patterns with patterns.

Solved with no hints

Stages solved without taking a hint

Compare similar puzzles

Answer shown

Stages where the last hint gave away the answer

Compare similar puzzles

Wrong answers

Wrong answers per player at each stage

Compare similar puzzles

Time to solve

The middle time for each stage and for the whole hunt. Half of players are faster and half are slower.

Compare similar puzzles

What we see	What it probably means	What we do
One puzzle is solved with no hints far more often than similar puzzles	It is too easy for these players	**Make that puzzle harder.**
One puzzle shows the answer far more often than similar puzzles	It is too hard or unfair	**Rewrite the puzzle or its hints. Remove it if it is still too hard.**
One puzzle gets many more wrong answers than similar puzzles, but no more hints	We may be rejecting right answers, or players may not see the hints	**Check which answers we accept, and make the hint button easier to see.**
More players finish, but fewer come back after 7 days	The puzzles may have become too easy to feel rewarding	**Bring back harder puzzles and check the mix.**

Is the mix of easy and hard puzzles right?

Planned: an uneven mix of easy, medium and hard puzzles

Players should not be able to guess how hard tomorrow's puzzle will be. So the mix will change from week to week, and it will not simply alternate. For example:

- First week of the month: 3 easy days, then 1 hard day.
- Next week: 2 hard days, 1 easy day, 1 hard day, then 3 medium days.

The game does not do this yet. Today the same seven hunts repeat in the same order, and they are not labelled by difficulty.

Solved with no hints, by difficulty

Stages solved without taking a hint, shown for easy, medium and hard days

Compare the 3 levels

Players who finish, by difficulty

Players who finish, out of players who pressed Start, shown for easy, medium and hard days

Compare the 3 levels

Back the next day, by difficulty

Players who finish a hunt and open the game the next day, shown after easy, medium and hard days

Compare the 3 levels

What we see	What it probably means	What we do
A puzzle labelled hard is solved with no hints as often as easy puzzles	The label is wrong	**Change the label, or make the puzzle harder.**
Far fewer players come back the day after a hard puzzle	Hard days are putting people off	**Make hard days a little easier, or follow them with an easier day.**
Fewer players come back during long runs of easy days	Too many easy days in a row feel boring	**Shorten the easy runs and bring in a hard day sooner.**

Does the streak keep people playing?

A streak counts the days in a row a player finishes a hunt. A freeze covers one missed day, once a week.

Players on a streak of 3 or more

Of the players who finished a hunt this week, those with a streak of at least 3 days

Set after two weeks

Back after a freeze

Players who finish a hunt within 2 days after a freeze covered a missed day

Compare with next row

Back after losing a streak

Players who finish a hunt on the day their streak reset, or the day after

Compare with row above

What we see	What it probably means	What we do
Players saved by a freeze come back much more often than players who lost their streak	The freeze may be keeping people playing	**Try allowing a second freeze, or a way to earn one.**
Both groups come back about as often	The freeze may not be worth having	**Make the streak rules simpler.**

Duolingo tested letting people hold two freezes instead of one, and its daily users rose by 0.38%. So any effect may be small, and we will need many players to see it.

Do players share it?

Players who share

Players who use Share or Copy, out of players who finish that day

Set after two weeks

What we see	What it probably means	What we do
Fewer players share than in our first two weeks	The result is not worth showing off	**Try a new share card and a challenge-a-friend link.**
Sharing stays steady, but new players do not grow	Shares are not bringing people in	**Track visits from shared links, then change the message or the link.**

Do players want more than one hunt a day?

Players who try practice hunts

Players who open a practice hunt, out of players who finish that day

Set after two weeks

Players who look back at past hunts

Players who view a past hunt's answers or play it again, out of players who open the game that day

Set after two weeks

What we see	What it probably means	What we do
More players try practice hunts than in our first two weeks	More people want more than one hunt a day	**Add more practice hunts or a weekly bonus hunt. Check that people still come back the next day.**
More players look back at past hunts than in our first two weeks	Players value their record of past hunts	**Make past hunts easier to find, for example with a monthly map.**

How findings reach management

Each Monday, management gets one short update with four parts:

1. Players who finished a hunt this week, compared with last week.
2. How many new players came back after 1 day and after 7 days.
3. The hardest and the easiest puzzle of the week.
4. One thing we learned, one decision we made, and what we will check next week.

TEMPLATE, TO FILL IN ONCE WE HAVE DATA

Players who finished a hunt this week: _____. Last week: _____.

New players who came back after 1 day: _____. After 7 days: _____.

Hardest puzzle: _____. Easiest puzzle: _____.

What we learned: _____. What we decided: _____. What we will check next week: _____.

Be careful when reading the numbers

Check how many players there are.

With 100 players, a result of 45% could really be anywhere from about 35% to 55%. Small differences mean little until each group has a few hundred players.

One player means one browser on one device.
Someone using a second device, a private window or cleared browser data counts as a new player. This adds extra new players and makes the come-back numbers look a little lower.

Streak numbers miss people who leave for good.
We only record a lost streak when a player opens the game. Someone who never returns records nothing, so read streak numbers next to the come-back numbers.

Freeze comparisons do not prove the freeze helped.
Players who use a freeze may already be keener. We need a proper test, where some players get the feature and some do not, before giving it credit.

Each puzzle always falls on the same weekday.
There are seven daily hunts on a seven-day cycle, so a hard puzzle and a quiet weekday look the same. The planned uneven mix, with more puzzles, would separate the two.

Puzzles repeat after seven days.
Returning players may remember the answers, so we judge how hard a puzzle is from first plays only.

Use the middle time, not the average.
The timer keeps running if a player leaves and comes back, so a few very long times would distort an average.

Copying is not the same as posting.
Copy shows the result was copied, not that it was posted anywhere.

The leaderboard is sample data.
Only the player's own row is real, so it tells us nothing about competition yet.

What we need before the numbers are real

Collect all players' data in one place

Today each player's data stays on their own device, so we cannot see totals.

Fix: Send the data to an analytics service. This comes first, because nothing else in this document works without it.

Label puzzles by difficulty

We cannot compare easy and hard days until each puzzle is labelled, and seven puzzles are too few for an uneven pattern.

Fix: Label each puzzle easy, medium or hard, write more puzzles, and schedule them in an uneven mix.

Tag shared links

We cannot tell whether shares bring in new players.

Fix: Add a tag to the link in the share text, and record it on a player's first visit.

Keep wrong answers

We cannot tell typos and fair alternative answers from real mistakes.

Fix: Record wrong answers in a way that protects privacy, and review the most common ones.

Learn why players stop

Leaving the page also happens when someone briefly switches apps, so it only shows where a player was.

Fix: Count "started but did not finish that day" instead, and add a one-question survey if needed.

Sources

CHECKED NOVEMBER 2026, NUMBERS ROUNDED

Typical puzzle game and top quarter: GameAnalytics, 2025 Mobile Gaming Benchmarks. Data from 2024, covering 11,600 iOS and Android games. Some top quarter numbers are read from charts. gameanalytics.com/reports/2025-mobile-gaming-benchmarks

50 most successful puzzle games: Gamigion, using AppMagic data. Middle values. The article does not say what year it covers or how the games were chosen. gamigion.com/benchmarks-from-the-top-50-mobile-puzzle-games/

Streak freeze: Duolingo, January 2022. Reported by the company and not independently checked. blog.duolingo.com/how-duolingo-streak-builds-habit/

Play the game: dailytrail.vercel.app

The same plan is kept with the game's code as docs/METRICS.md. Exact technical definitions are in reference/events.md.